Plan 08: Learned Memory Wrapper for a Frozen-Base LLM

Research Progress / Fact Record

Mingjia Shi

Long-Context Memory / Frozen-Base Adaptation

June 5, 2026

Research progress / fact record

Weekly inputs are appended below in chronological order.

Reporting Period: May 2026 Week 4

Date range: Mon 2026-05-25 to Sun 2026-05-31
Month/week is determined by the Monday of the reporting week.

Background

- RCA and debugging tasks are becoming long-context problems: logs, traces, stack traces, test failures, code context, topology, and prior observations often exceed a practical prompt budget.
- Direct long-context prompting is expensive and brittle: high prefill cost, large KV cache, lost-in-the-middle behavior, and no persistent learning between chunks.
- Directly fine-tuning an 8B base on our small RCA/debug data is risky because we do not have the original pretraining mix needed to protect the base model's general capability.
- This project starts from a frozen 7B/8B open-source base and trains a separate wrapper memory module.

Significance

- **Research value**: formulate long-context compression as next-state prediction over learned memory states:

$$m_t = G_\phi(m_{t-1}, c_t), \qquad y_t = F_\theta(q_t, m_t)$$

- **Product value**: build an RCA foundation model for structured root-cause analysis, with lower context cost and better evidence retention.
- **Risk control**: train only the wrapper in v0, keeping the base model frozen to reduce catastrophic forgetting.
- **Community impact**: release a model, benchmark protocol, demo, and technical report rather than only internal experiment logs.

Progress / Fact Record

- **Done**: scoped Plan 08 v0 as a learned-memory wrapper project rather than full self-modifying model training.
- **Done**: read the `rca-demo` SFT results as project initialization: SFT can teach RCA signals, but it is not always a win and is not enough for robust RCA foundation-model capability.
- **Done**: defined two deliverable tracks:
  1. Research paper: new architecture for memory-state next prediction.
  2. RCA foundation model: public RCA datasets first, with public debug traces as a later extension.
- **Done**: added English and Chinese notes plus a dedicated v0 budget.
- **Done**: created the progress/fact-record slide structure.
- **Not verified yet**: no wrapper training result, LoCoMo result, public debug-trace result, or compression demo metric has landed yet.

Status Legend

- **Done**: implemented, documented, or measured from existing artifacts.
- **Verified**: backed by a completed run, metric, or reproducible output.
- **Planned**: design direction or future milestone, not necessarily immediate.
- **TODO**: near-term action to execute next; should be actionable this week / next week.
- This deck separates project facts from plans so readers do not confuse proposals with results.

Status Summary

Done

- Plan 08 v0 scoped.
- RCA-demo SFT boundary read.
- Research/project tracks defined.
- EN/ZH notes and v0 budget written.
- Slide structure created.
- Demo scripts identified.

Planned

- Frozen 8B base + wrapper path.
- LoCoMo / LongBench scaling.
- RCA foundation-model path.
- Trainable wrapper memory.
- Research paper after validation.

TODO

- Select first demo case.
- Generate four prompt variants.
- Run $q25_{base}$ / $q25_{mix}$.
- Score demo outputs.
- Package comparison table.

RCA-Demo Initialization: SFT Boundary Search

- Goal: use `rca-demo` to find how far standard SFT can push a general model toward RCA behavior.
- Setup:
  - Displayed base family: Qwen2.5-7B-Instruct (q25).
  - 4 RCA datasets: Nezha, OpenRCA-500, RCAEval, lincyaw/rca.
  - Training schedules: base, curriculum chain, and joint mix.
  - Evaluation matrix: RCA metrics + general capability benchmarks.
- Project role: this is not the final solution; it is the initialization phase that maps the SFT capability boundary.

SFT Wins: The Model Can Learn RCA Signals

- $q25_{mix}$ became the strongest complete row by `recommendation_f1`:
  - OpenRCA-500: 0.305
  - RCAEval: 0.733
  - lincyaw: 0.242
- After base-mismatch re-eval, $q25_{curr\_4\_lincyaw}$ showed real cumulative curriculum signal:
  - lincyaw `service_set_f1`: 0.272 pre-fix $\rightarrow$ 0.563 post-fix.
  - lincyaw `primary_service_match`: 0.454 $\rightarrow$ 0.723.
  - RCAEval `severity_match`: 0.041 $\rightarrow$ 0.818.
- Takeaway: SFT is useful for injecting RCA structure, service identification, and recommendation behavior.

SFT Boundaries: Not Always a Win

- General benchmark regression appears after direct RCA SFT:
  - $q25_{mix}$ GSM8K drops $0.795 \rightarrow 0.715$.
  - $q25_{mix}$ TruthfulQA drops $0.647 \rightarrow 0.589$.
- Takeaway: SFT can teach RCA behavior, but general-capability regression must be tracked before scaling.

Why Wrapper Memory Follows From RCA-Demo

- Direct base-model SFT exposes three project risks:
  1. capability drift without the original pretraining distribution;
  2. output-format fragility under LoRA updates;
  3. evidence-grounding loss even when JSON surface form improves.
- Frozen base + wrapper separates concerns:
  - base model keeps general reasoning and coding capability;
  - wrapper absorbs long context, tool observations, and RCA/debug traces;
  - failures can be isolated to memory compression instead of the whole model.
- This turns `rca-demo` from a final SFT path into evidence for the RCA foundation-model architecture choice.

First Delivery: A Demo, Not the Full System

- **Planned**: first delivery is a small, convincing demo, not a full paper-quality model release.
- The demo is intended to test the direction:
  1. long context is expensive / brittle;
  2. simple SFT has a capability boundary;
  3. frozen base + wrapper memory is a plausible next step;
  4. the same incident can be compared across full context, summary, retrieval, and memory.
- **TODO**: produce one reproducible case, one comparison table, and one qualitative walkthrough as the next delivery.

Demo Segment 1: Problem Setup

- **TODO**: pick one RCA/debug case with enough evidence to be genuinely long-context.
- Show the raw context shape:
  - issue or incident description;
  - logs / traces / failing test output;
  - code or service context;
  - distractor evidence.
- **Not verified yet**: we have not selected the demo case or measured its token/KV footprint.
- Management question: can the model identify root cause without reading everything directly?

Demo Segment 2: Baseline Comparison

- **TODO**: run the frozen 8B base with three non-learning strategies:
  1. full context;
  2. fixed-budget summary;
  3. retrieval top-k chunks.
- **TODO**: measure:
  - answer quality and evidence grounding;
  - prompt tokens / estimated KV cost;
  - format stability.
- **Not verified yet**: no full-context / summary / retrieval result has been run for the demo case.

Demo Segment 3: Wrapper Memory Path

- **Planned**: process the same context as a chunk stream:

$$m_t = G_\phi(m_{t-1}, c_t)$$

- **Planned**: generate the RCA/debug answer from compressed memory:

$$y_t = F_\theta(q_t, m_t)$$

- **TODO**: use deterministic memory compression for the immediate demo path.
- **Planned**: replace deterministic memory with trained explicit memory tokens after the demo.
- **Not verified yet**: no trained wrapper output has been measured.
- Constraint: keep the base model frozen; all adaptation happens in the wrapper.

Demo Segment 4: What We Need to Show

- The demo does not need to beat every benchmark yet.
- **Target, not verified**: it needs to show a credible direction:
  - lower prompt/KV cost than full context;
  - better evidence retention than naive summary;
  - more complete causal reasoning than retrieval-only;
  - no base-model finetuning required.
- **Decision gate**: only scale to LoCoMo / LongBench / public debug traces after the demo beats cheap baselines on at least one meaningful axis.

Demo Segment 5: Deliverable Package

- **TODO**: demo script that builds one long-context case and produces four prompts.
- **TODO**: output table comparing full context, summary, retrieval, and wrapper memory.
- **TODO**: qualitative page showing where each strategy succeeds or fails.
- **TODO**: short technical note explaining whether the result motivates wrapper training.
- **Decision**: continue to wrapper training only if the demo beats the cheap baselines on at least one meaningful axis.

Demo Assets Already Available in RCA-Demo

- **Done**: `rca-chat` Gradio probe exists and can load matrix checkpoints:
  - $q25_{base}$, $q25_{mix}$, $q25_{curr\_*}$;
  - public RCA datasets plus optional private Liang DTS bundle.
- **Done**: long-context builder script exists:
  - `scripts/compression/build_long_context_rca_benchmark.py`;
  - converts prepared `cases.jsonl` into chunked RCA items with distractors.
- **Done**: prompt-strategy preparation script exists:
  - `scripts/compression/prepare_compression_eval.py`;
  - emits `full_context`, `summary`, `retrieval`, and `learned_memory` prompt rows.
- **Not verified yet**: no end-to-end generation/score run has been completed for this demo path.

Demo Case Selection

- **Preferred public demo**: one RCAEval case.
  - It has microservice fault-injection structure.
  - It supports service/fault-type metrics.
  - It is public and suitable for release.
- **Alternative public demo**: one OpenRCA-500 case.
  - Wider service vocabulary.
  - Good for showing service-level root-cause ranking.
- **Internal-only qualitative demo**: Liang / Nokia DTS.
  - Useful for realism.
  - Not releasable; do not include raw prompts or per-case outputs in public slides.

Demo Build Commands

Step 1: build long-context RCA items

```
python3 scripts/compression/build_long_context_rca_benchmark.py \
  --cases runs/rcaeval_v1/prepared/cases.jsonl \
  --out runs/long_context_rca_demo/items.jsonl \
  --chunk-chars 1800 \
  --overlap-chars 150 \
  --distractors 2
```

Step 2: prepare four prompt strategies

```
python3 scripts/compression/prepare_compression_eval.py \
  --input runs/long_context_rca_demo/items.jsonl \
  --out runs/long_context_rca_demo/prompts.jsonl \
  --summary-chunks 8 \
  --retrieval-k 8 \
  --memory-records 8
```

Demo Four-Way Comparison

Strategy	What it uses	What we learn
Full context	All chunks in prompt	Upper-bound quality, high token/KV cost.
Summary	Fixed-budget compressed text	Cheap baseline; may lose evidence details.
Retrieval	Top-k salient chunks	Good for evidence lookup; may miss causal synthesis.
Learned memory	Rendered wrapper memory state	Whether memory can retain evidence at lower context cost.

Not verified yet: the current `learned_memory` row is a deterministic memory-compressor baseline until the trainable wrapper is run.

Demo Output Layout and Metrics

- **Planned output layout** mirrors matrix cells:
  - `runs/long_context_rca_demo/eval/<model_id>/<strategy>/predictions.jsonl`
  - `metrics.json`
  - `case_scores.jsonl`
  - `eval_command.txt`
- **Metrics to report**:
  - RCA quality: `recommendation_f1`, `root_cause_f1`, `primary_service_match`, `service_set_f1`;
  - grounding: `evidence_overlap`;
  - format: `json_parseable`, `json_repaired`, `degenerate`;
  - compression: prompt token estimate, compression ratio, latency if available.
- **TODO**: wire `prompts.jsonl` into generation and scoring.

Demo Model Choice

- **Baseline model for first demo**: $q25_{mix}$.
  - Existing matrix says it is the strongest complete RCA row by `recommendation_f1`.
  - RCAEval `recommendation_f1` $= 0.733$.
  - JSON format is stable in the public RCA matrix.
- **Control model**: $q25_{base}$.
  - Shows how much RCA behavior came from SFT.

Core Architecture

Memory update

$$m_t = G_\phi(m_{t-1}, c_t)$$

$$y_t = F_\theta(q_t, m_t)$$

- F_θ: frozen 7B/8B base model.
- G_ϕ: trainable wrapper.
- m_t: compact memory state.
- c_t: new context chunk or tool observation.

V0 memory type

Start with explicit memory tokens.

- Easy to train and ablate.
- Compatible with frozen decoder models.
- Safer than writing into weights.
- Hidden-state memory and LoRA memory are later ablations.

Why Freeze the 8B Base?

- We start around 8B because it is strong enough for reasoning and still feasible for repeated experiments.
- Candidate bases:
  - Llama-3.1-8B-Instruct as a secondary validation base.
  - Qwen2.5-Coder-7B or similar for public debug-trace extension.
- The base remains frozen in v0.
- Training only the wrapper limits catastrophic forgetting and makes the contribution easier to isolate.

Two Tracks

Track A: Research Paper

- New memory-wrapper architecture.
- Next-state prediction over memory states.
- Evaluate on long-context memory benchmarks.
- Main claim: compact learned memory can replace part of long prompt context.

Track B: RCA Foundation Model

- Public RCA model with internal qualitative probe support.
- Use code issues, stack traces, failing tests, patches.
- Direct prediction and agentic tool-use modes.
- Release model, report, and demo.

Dataset Plan

Track	Datasets	Purpose
Base long-context	LoCoMo, LongBench, RULER, Needle-in-a-Haystack, Qasper, NarrativeQA	Prove wrapper memory before domain-specific RCA.
Public debug traces	SWE-bench Verified, BugsInPy, Defects4J, QuixBugs, curated CodeNet/APPS traces	Later extension for debugging and trace-based root-cause reasoning.
RCA domain	Nezha, OpenRCA-500, RCAEval, lincyaw/rca	Transfer compression method to RCA evidence bundles.
Internal probe	Liang / Nokia DTS	Qualitative validation only; not for public release.

Evaluation Modes

Direct prediction

Input: issue text, code context, stack trace, logs, failing test output.
Output: root cause, evidence, fix recommendation.

Agentic RCA

The model can call tools such as search, open file, run tests, and inspect traces.
Test-time training updates the wrapper from tool observations; the frozen base does not change.

Baseline comparison

Full context vs. summary vs. retrieval vs. learned memory.

Success Metrics

Quality

- RCA / debug root-cause accuracy.
- Fix recommendation quality.
- Evidence grounding.
- JSON / schema stability for RCA outputs.

Compression

- Token and KV-cache reduction.
- Latency reduction.
- Early / middle / late evidence retention.
- Robustness to chunk order and distractors.

Budget Estimate

Scope	Time	GPUh	Cost	Output
Lean MVP	4–7 wks	1,420–2,950	\$4.3k–\$8.9k	One wrapper, smoke + RCA demo.
Paper-quality v0	7–13 wks	2,820–5,750	\$8.5k–\$17.3k	Long-context + RCA foundation model + ablations.
Strong release	10–16 wks	5,000–9,000	\$15k–\$27k	Multiple seeds, model card, polished report.

Cost model: H100 at \$3 per GPU-hour.

Execution Timeline

Phase	Duration	Decision gate
0. Smoke	3–5 days	Wrapper beats same-length summary on synthetic / NIAH.
1. Long-context	2–3 weeks	Beats summary on LoCoMo / LongBench / RULER.
2. Public debug traces	2–4 weeks	Improves direct root-cause / fix prediction on public debug traces.
3. RCA domain	1–2 weeks	Works on public RCA datasets with at least 4x compression.
4. Release polish	1–2 weeks	Ablations, demo, model card, technical report.

Current Status

- Plan 08 v0 has been scoped as a learned-memory wrapper project.
- English and Chinese notes are maintained in the research notes repo.
- Budget is separated from the full self-modifying-LLM version.
- RCA foundation-model path is defined around public RCA datasets first, with public coding/debug data as a later extension.
- Initial RCA compression scripts and report templates are drafted in the RCA demo repo.
- RCA-demo SFT initialization has identified the boundary: useful RCA learning exists, but direct SFT is not robust enough as the final RCA foundation-model route.

Next Steps

1. Build the first demo case and four prompt strategies.
2. Run frozen 8B base baselines: full context, summary, retrieval.
3. Add wrapper-memory path for the same case.
4. Package the demo table and qualitative walkthrough.
5. After demo validation, scale to LoCoMo / LongBench and public debug-trace datasets.

Ask / Discussion

- Is frozen-base + wrapper the right first milestone?
- Which 8B base should be the first public model target?
- Should the first paper emphasize long-context memory, RCA foundation modeling, or both?
- Which demo case would be most convincing for external readers?

Reporting Period: June 2026 Week 1

Date range: Mon 2026-06-01 to Sun 2026-06-07

Weekly meeting logic: why this matters → what we are doing → goal → progress.

Updates compiled on 2026-06-04/05 belong to this same Week 1 report.

Weekly Recall: Why Are We Doing This?

Problem

RCA/debugging needs long evidence: logs, traces, topology, prior cases, and code context. Directly fine-tuning the base model can teach RCA behavior, but may weaken general reasoning or retrieval ability.

- We want domain adaptation without overwriting the frozen base model.
- The useful module should compress long context, be reusable, and be safe to turn off when it is not appropriate.
- This connects May Week 4 to June Week 1: first build a wrapper, then measure when it is safe to use.

What Is The Use?

Practical use

A memory wrapper could make long-context RCA cheaper and more modular: keep the base model frozen, attach a small learned memory path, and reuse compressed evidence.

- Avoid full-context cost when gist-level memory is enough.
- Preserve a fallback path to the original base model.

Research use

It gives a clean testbed for frozen-base adaptation: what can a fixed-size soft memory encode, and when does it fail?

- Positive result: wrapper can learn.
- Boundary result: fixed memory has a capacity wall.

What Are We Doing?

1. **Train a wrapper** around a frozen base model: long evidence context $c_{1:n}$ is compressed into soft memory m.
2. **Evaluate the wrapper** against no-context, full-context, retrieval, and Gist-style baselines.
3. **Characterize the boundary**: where memory helps, where it is neutral, and where it collapses.
4. **Add a gate**: only let memory participate when intrinsic signals suggest compression is safe.

Short version

We are not claiming "memory replaces context." We are building controlled frozen-base adaptation: learn memory, measure its limits, then gate it.

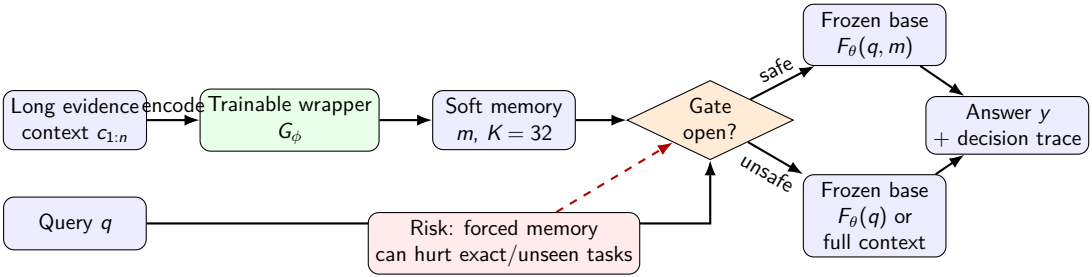
Goal For This Line Of Work

Main goal

A frozen-base memory system that improves long-context/RCA cases where compression is useful, while preserving the base model when memory is unsafe.

- **v1**: characterize the learned soft-prompt wrapper; write the result as a bit-capacity-wall paper.
- **v1.5**: build a do-no-harm gate between wrapper and base.
- **v2**: extend from one-shot compressed memory to cross-session latent memory with read/write behavior.

Architecture: Input, Memory, Gate, Output



How Is It Going? v1 Result

Benchmark	Wrapper	Gist	Best reference
QuALITY	0.193 ± 0.032	0.180 ± 0.044	no-context 0.141
MuSR-mm	0.493 ± 0.008	0.501 ± 0.013	full-context 0.551
RULER-NIAH	0.000 ± 0.000	0.000 ± 0.000	full-context 0.995

- **Good news**: the wrapper can train and helps on compressible gist-style inputs.
- **Lesson**: it is not a universal replacement for long context; exact-retrieval cases expose a capacity wall.
- **Setting**: P08-S2; full facts in v1 results summary.

How Is It Going? Interpretation

1. **Wrapper can learn**: memory is useful when the answer can be represented as compressed/gist information.
2. **Wrapper changes inference**: it is not neutral; the learned memory path can pull the base away from safer behavior.
3. **Always-on memory is risky**: if the task needs exact details or unseen reasoning transfer, forced memory can lose information.
4. **Next design**: put a gate between wrapper and base, and open it only when intrinsic signals say compression is safe.

How Is It Going? v1.5 Gate

Signal family	Interpretation	Direction	Use
Confidence / seq logprob	wrapper likely correct	positive	open
Wrapper-to-base KL / JS / TV	wrapper fights base	inverse	close
Memory/write dynamics	wrapper competence signal	positive	open/close
Mid/late-layer drift	late reasoning distorted	inverse	close

- **Goal**: keep in-distribution gains while avoiding OOD harm.
- **Setting**: P08-S3/P08-S6 for signal probes; detailed v1.5 results continue in the appended gate section.
- Main deliverables: signal grids and v1.5 PDF.

This Week's Status Summary

Done

- v1 reframed as bit-capacity characterization.
- Main result cells and settings are linked.
- Gate direction is now concrete, not vague.

Next

- Polish v1 paper story.
- Run/validate do-no-harm gate.
- Test if multi-task training widens the safe region.

Meeting takeaway

The project is moving from "can memory learn?" to "can memory be used safely?" The current answer is positive but bounded: useful compression exists, and the gate is the mechanism that can make it deployable.

Where To Read More

Item	Link
Plan 08 hub	README / deliverables hub
v1 facts	main results summary (setting P08-S2)
v1.5 report	compiled PDF (setting P08-S3)
v1.5 signal grid	CSV + figures (setting P08-S3)
v2 setting	v2 plan (setting P08-S4)

v1.5: can a learned memory module be deployed *safely*?

- A frozen base + a small learned **memory wrapper**.
- **The question:** can we get its gains *where it works* and *never lose points* where it doesn't?

One-line answer: **yes** — an intrinsic-signal gate makes the wrapper *help in-distribution* and *do no harm out-of-distribution*, and it **transfers across 7 model families with no tuning**.

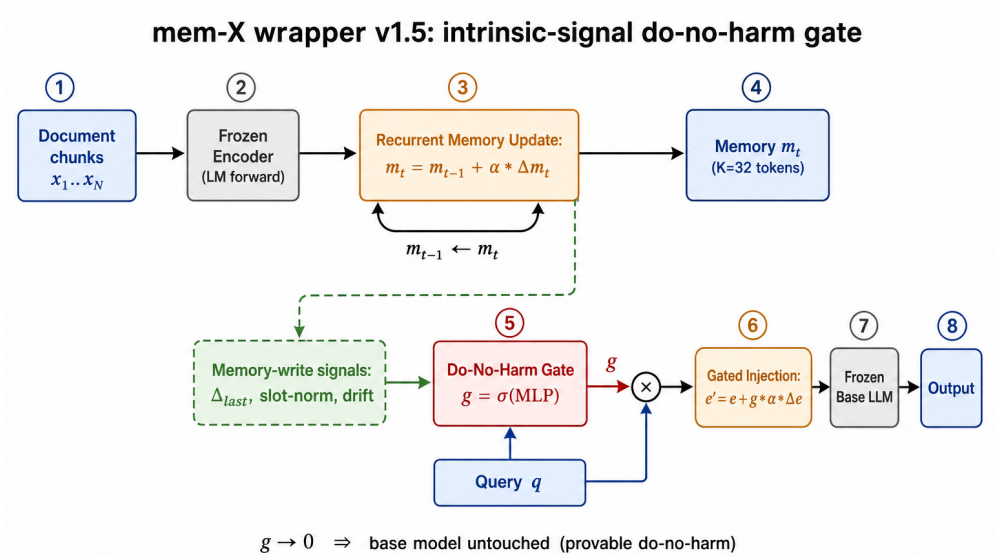
→ every term/number on the next slides links out to a wiki page: `v1.5-glossary.md`.

Why it matters

- **Frozen base = a promise: don't regress.** A memory/adaptation module is only deployable if it can't quietly *hurt* the model on inputs it wasn't built for.
- **Our v1 finding:** the memory wrapper is a *narrow lossy compressor* — it helps on some inputs and **actively hurts** on others (e.g. exact-answer QA). Negative transfer is the blocker.
- **So the real product isn't "a better wrapper"** — it's **knowing when to use it**, automatically, on any base model.

→ v1 characterization (3-regime law): `v1-results-2026-06-03.md`.

What we do (architecture)

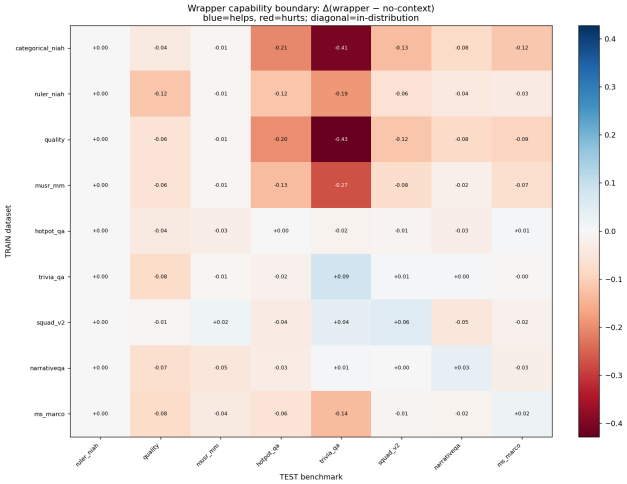


The bet — why this should work

- **Hypothesis:** whether the wrapper will help is *readable from the model's own forward pass* — specifically from how much it **wrote into memory** ($\|\Delta m\|$, slot norms, drift).
- If true, a tiny gate on those signals can decide **per input** whether to open the wrapper or fall back to the base.
- **Crucial requirement:** those signals must be *the same across model families* — otherwise we'd re-tune per model and it's not deployable.

→ candidate signals & screen: `v1.5-metric-candidates-2026-06-04.md`; setting `P08-S6`.

Result 1 — *where* it helps (expected vs. real)



Expected: helps where trained, decays as test drifts from train.

Real (72 train×test cells):

- **in-distribution: it helps** (QA +2 to +9 pt).
- competence is **distribution-bound**: gain → harm already at the same-task line.

→ P08–S7, matrix §8.

Result 2 — *deploy it safely* (expected vs. real)

Expected: one gate keeps the in-dist gains, avoids the OOD harm, and works on a new base model unchanged.

Real

- **do-no-harm holds**: gated $\geq$ base on **32/35** cases.
- **model-agnostic**: the gate signal (`delta_last`) predicts usefulness on an **unseen** family at AUROC **0.71** (leave-one-model-out, 7 families).
- **which signal generalizes:** memory-write dynamics (yes) vs. logit-lens confidence (no — model-specific).

$\rightarrow$ `P08-S6/P08-S8`, matrix §2b/§7; soft-gate that *failed* (and why) = `v1.5-gate-sweep-2026-06-05.md`.

So what + honest scope + next

Takeaway: a frozen-base memory module becomes **safely deployable** — it captures gains where it's competent and is provably harmless elsewhere, using one *model-general* intrinsic signal.

- **Honest scope:** in-dist gains are *modest & dataset-dependent* (QA yes, multiple-choice no); the headline value is **safe, modular memory**, not SOTA.
- **Next:** (1) *locking experiment* — one gate that opens in-dist & closes OOD on the same wrappers; (2) significance (more seeds + CIs); (3) *mix-train* (all datasets) — does multi-task training widen the boundary?

→ full ledger & asset index: `mem-embedding/summary/matrix.md`; decode any term: `v1.5-glossary.md`.